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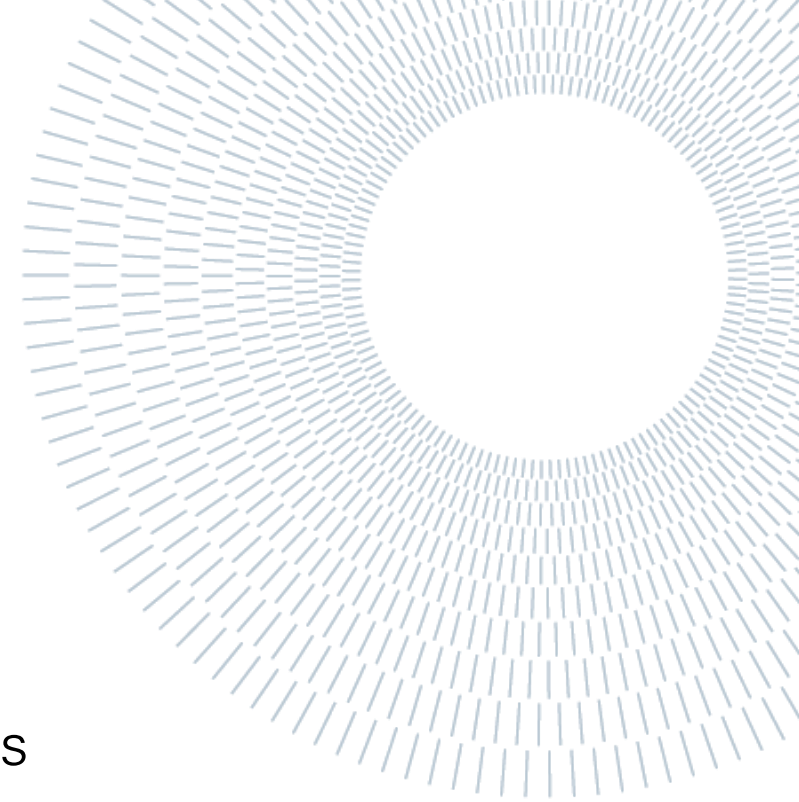
SPACECRAFT ATTITUDE DYNAMICS AND CONTROL

Lab 2.2

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A.Y. 2025-2026

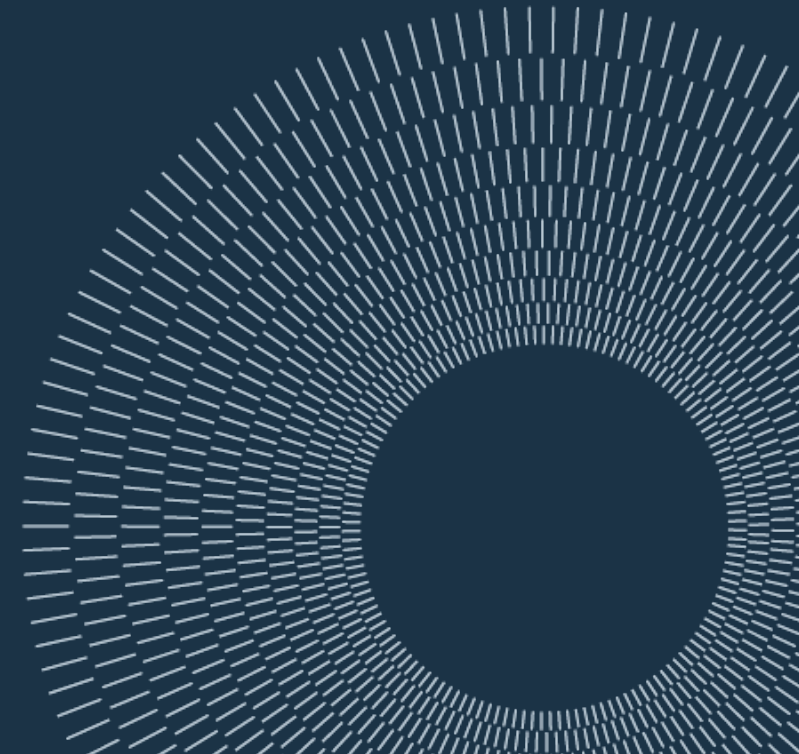
- Simulink-MATLAB Interface
- Exercises
- Real Vs Discrete World
- Debug – Order of Operations



Spring-Mass-Dumper

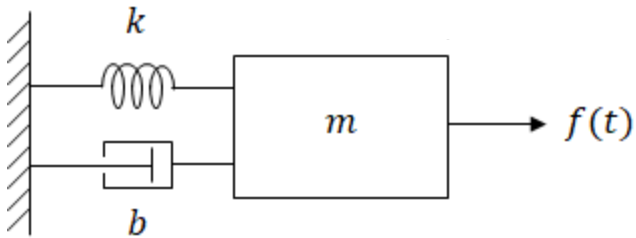


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Spring-Mass-Damper

We want to simulate the motion of a spring-mass-damper system under the effect of a force $f(t) = \sin(\omega t)$. We want to visualize both the position x and the velocity $v = \dot{x}$.



$$m\ddot{x} + b\dot{x} + kx = \sin(\omega t)$$

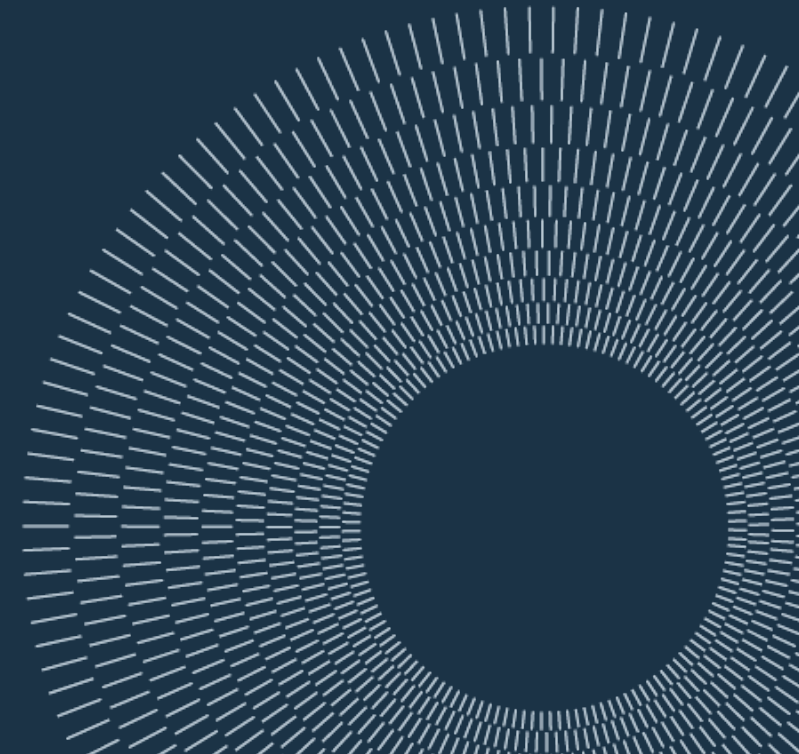
$$m = 2, b = 5, k = 6, \omega = 0.5$$

$$x_0 = 1, v_0 = 0$$

Simulink-MATLAB Interface



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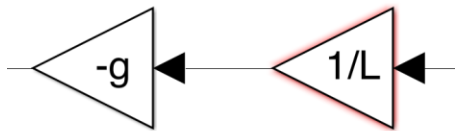


Interactions with MATLAB

Simulink can interact with MATLAB in different ways. The most useful ways regard the sharing of the workspace.

Access to MATLAB workspace

Simulink can access MATLAB variables that are present in the current workspace.

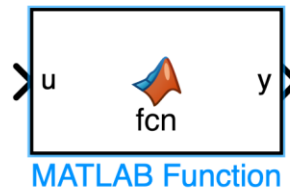


(dangerous!)

(great if using caller script!)

MATLAB Functions

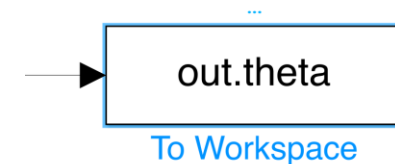
Simulink can execute MATLAB functions using the MATLAB Function block



(not suggested!)

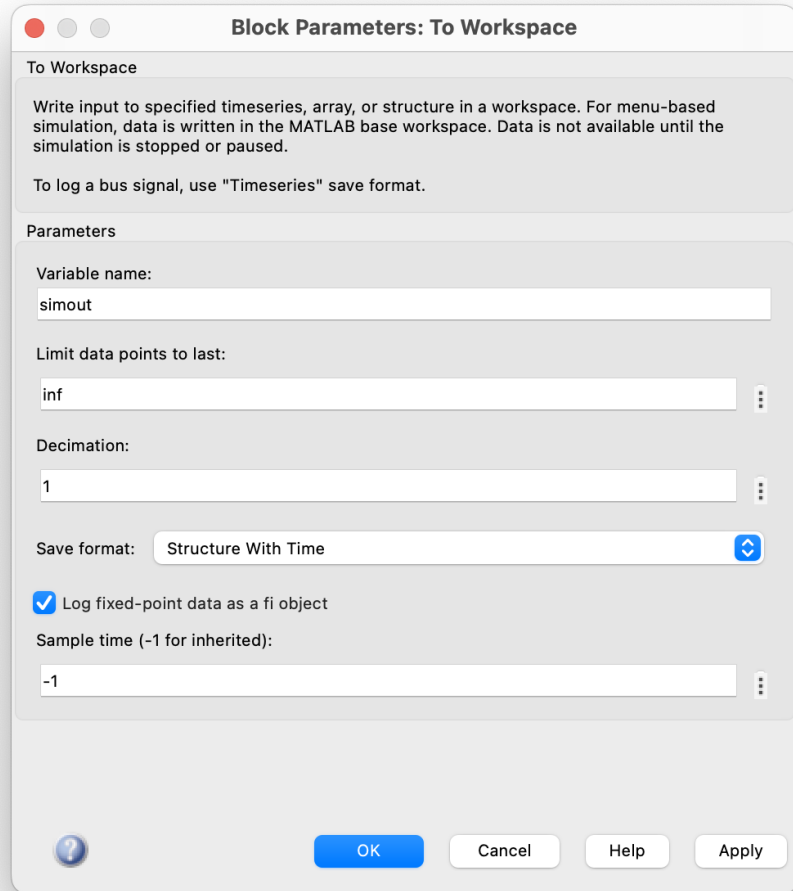
To/From Workspace

Simulink variables can be exported to MATLAB using the "To Workspace" block.



(perfectly fine)

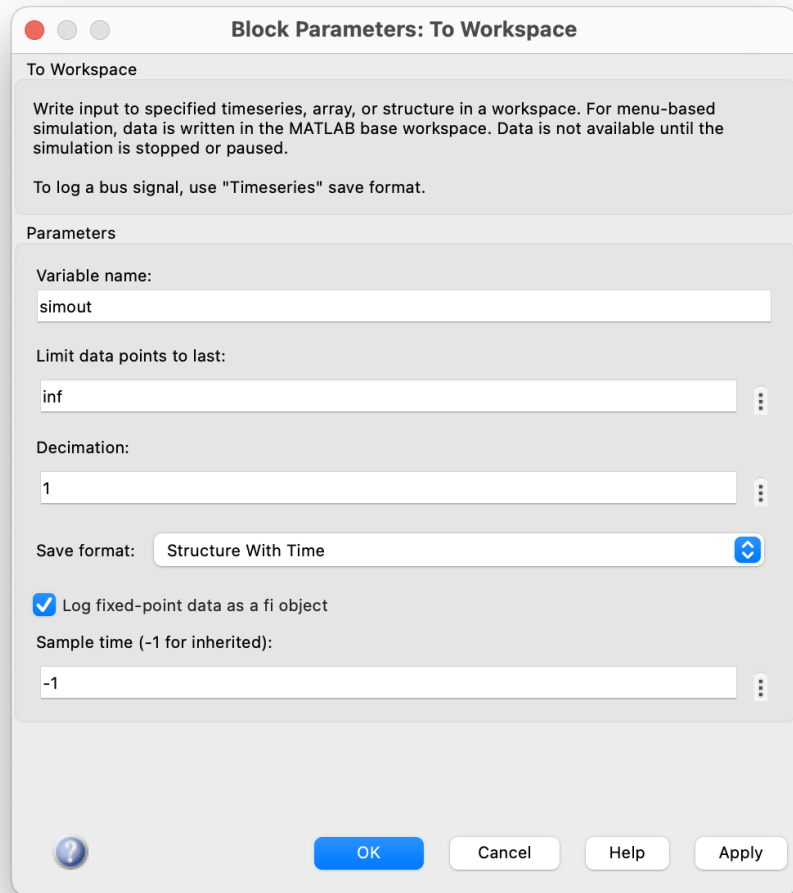
Common Blocks Settings | To Workspace



The *To Workspace* block allows transferring the simulation outputs to MATLAB.

- **Variable name:** the field name of the `out` structure associated to the signal in the MATLAB workspace.
- **Limit data points to last:** we can choose to save the last N data points only.
- **Decimation:** we can choose to save the data at each N timesteps

Common Blocks Settings | To Workspace

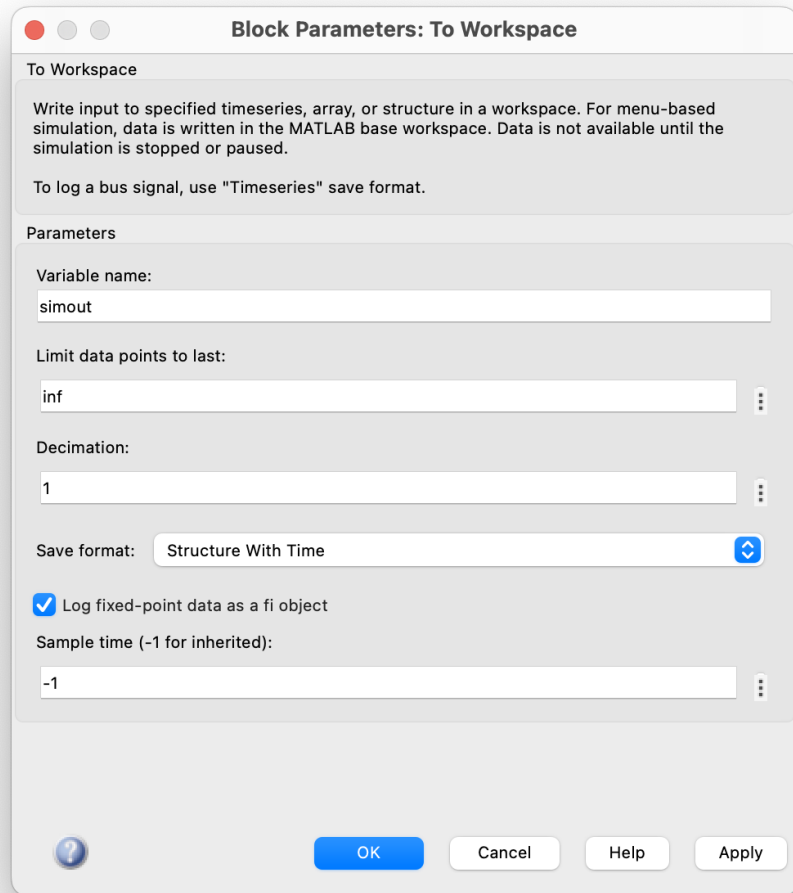


- **Save format:** the signal will appear in the MATLAB workspace as:
 - *Structure with time:* a structure with the fields **time** and **signals**. The **signals** sub-field contains the field values, dimensions, and label

```
out
|-- simout
    |-- time
    |-- signals
        |-- values
        |-- dimensions
        |-- label
```

- *Structure:* same as before, but the time array is empty

Common Blocks Settings | To Workspace

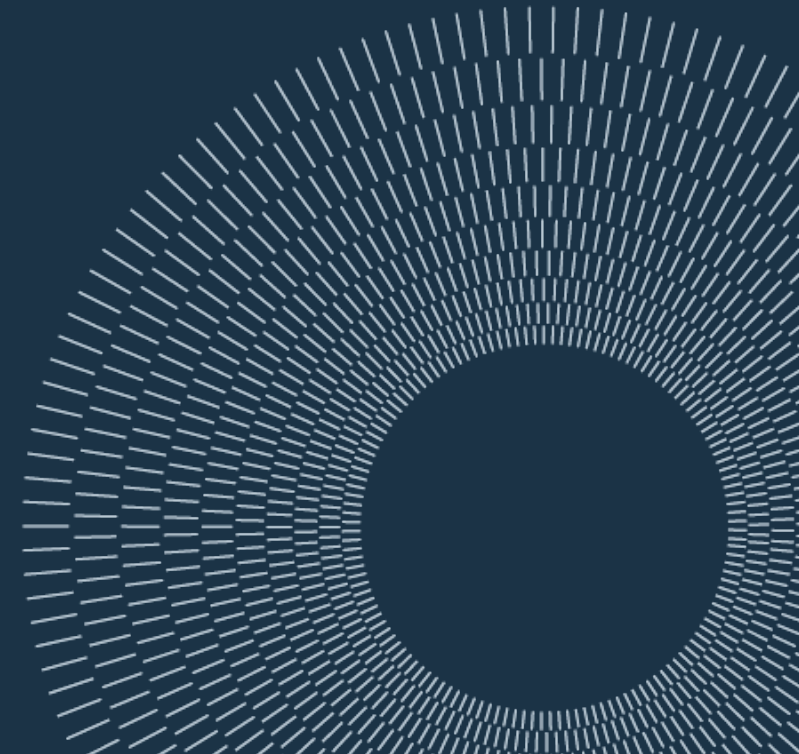


- **Save format:** the signal will appear in the MATLAB workspace as:
 - *Array*: the raw data signal, with no time information
 - *Timeseries*: a complex MATLAB class that contains additional information on the signal
- **Log fixed point data as fi object:** a setting specific for fixed-point arithmetics. MATLAB usually uses floating-point arithmetics.
- **Sample time:** if we want this block to be sampled at a specific sample time. If inherited, it inherits the sample time of the input signal.

Exercise



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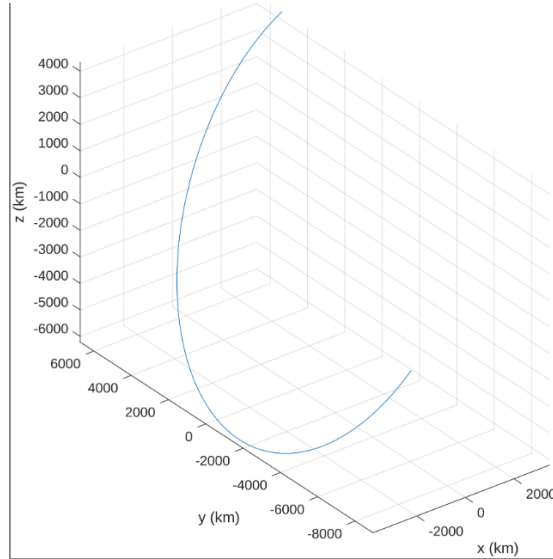
1. Compute the norm of a vector

Compute the norm of a vector in Simulink and check the result with Matlab.

$$||\mathbf{r}|| = \sqrt{r_x^2 + r_y^2 + r_z^2}$$

Note: it doesn't exist a «norm» block in Simulink.

2. Two-body Dynamics



We want to simulate the dynamics of a spacecraft influenced by only the acceleration of the main attractor for 1 day. We want to visualize the position x in a 3D plot.

$$\ddot{\mathbf{r}} = -\mu_{\text{Earth}} \frac{\mathbf{r}}{\|\mathbf{r}\|^3}$$

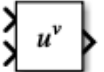
$$\mu_{\text{Earth}} = 3.986 \times 10^5 \frac{\text{km}^3}{\text{s}^2}$$

$$\mathbf{r}_0 = \begin{bmatrix} 3.46 \times 10^3 \\ 5.36 \times 10^3 \\ 4.34 \times 10^3 \end{bmatrix} \text{ km}$$

$$\mathbf{v}_0 = \begin{bmatrix} -2.27 \\ 4.05 \\ -5.81 \end{bmatrix} \frac{\text{km}}{\text{s}}$$

$$t_0 = 0, \quad t_f = 1 \text{ day}$$

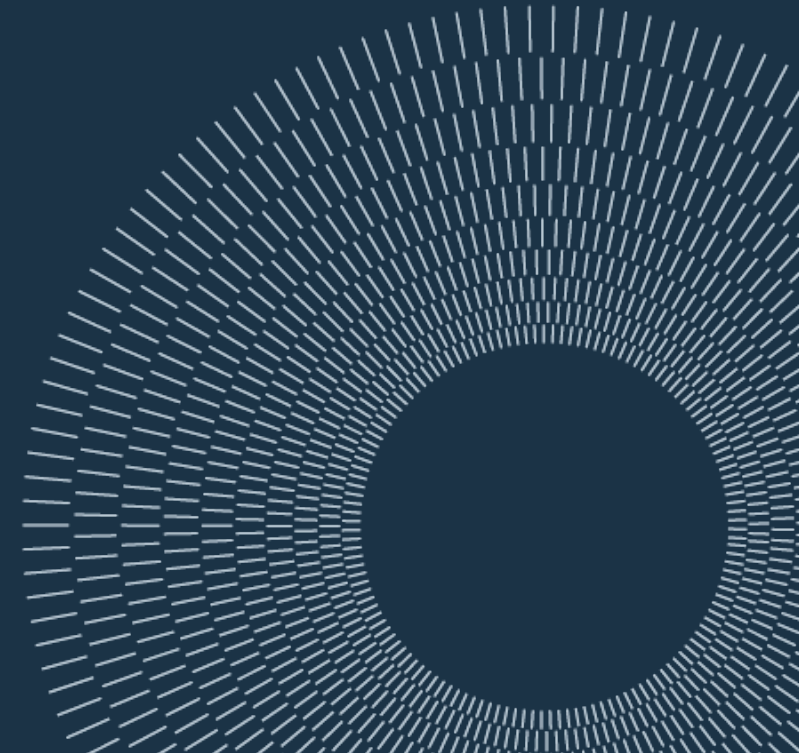
Useful boxes:

- Power 
- Norm (defined before)

Real vs Discrete Worlds



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Real World vs Discrete World

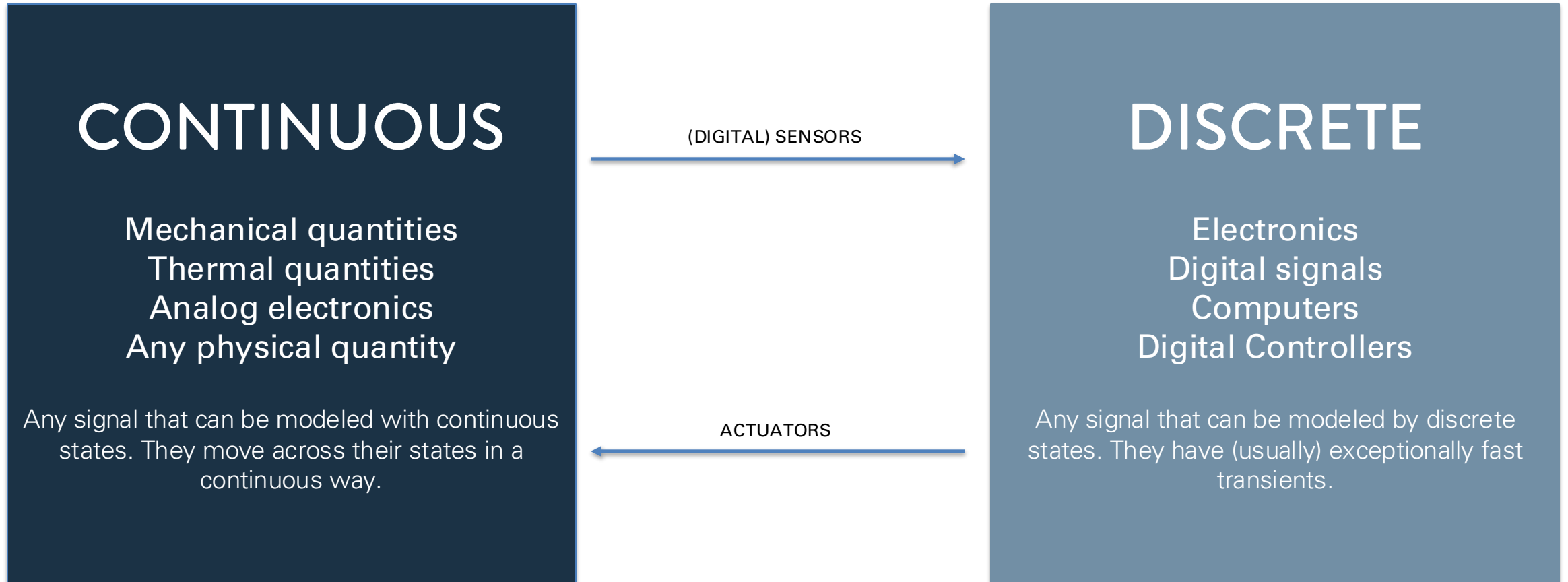


- Computers work on discrete clocks
- Every controller is a computer
- Every sensor is a computer

Before modeling anything, ask yourself:

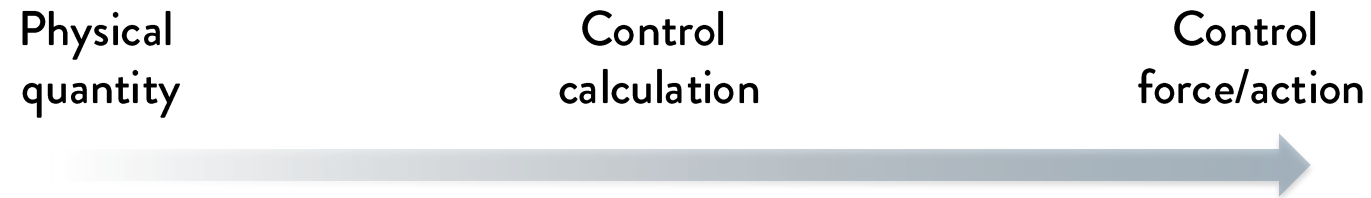
«Is this going to work in continuous or discrete time?»

Real World vs Discrete World

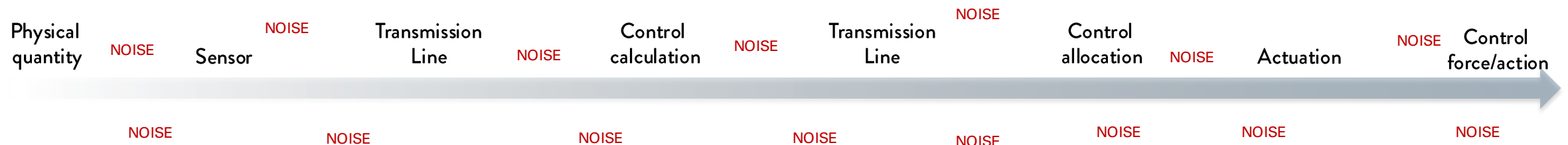


Ideal Systems vs Real-life Systems

Ideal system:



Real-world system:

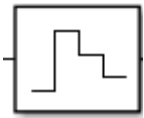


Discrete Worlds - Sensors

The continuous signal is sensed by a physical sensor.

Sensors can be seen as converters from the continuous to the discrete world.

To model the behaviour of the sensor, you should introduce (for example):



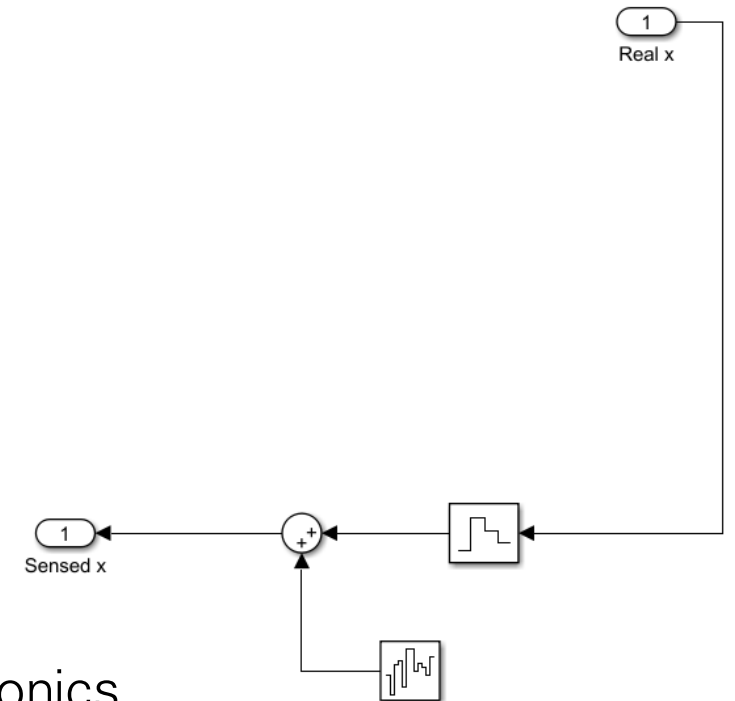
Zero-order hold block

mimics the sampling acquisition frequency of the sensor



White-noise block

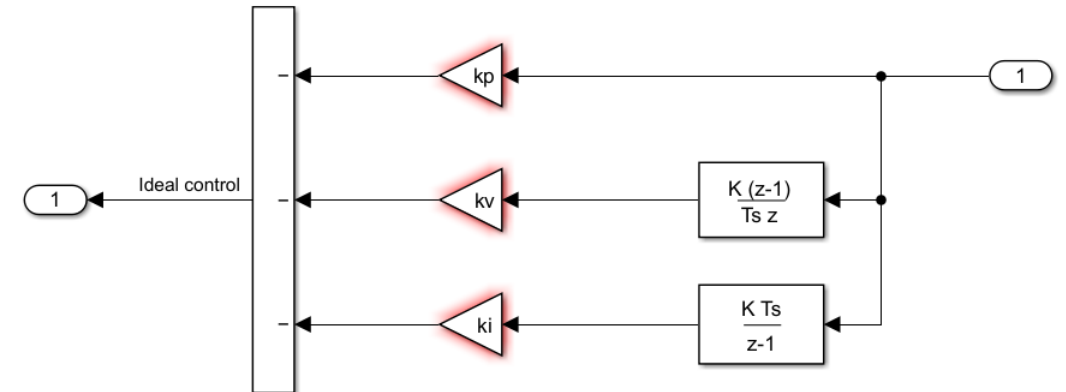
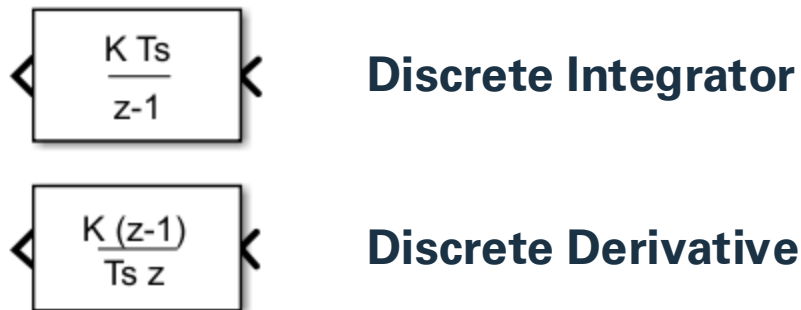
represents errors in the values read by the sensors' electronics



Discrete Worlds - Controller

The sensed quantity is input to a physical PID, which will integrate and derive the sensed quantity in a discrete way.

At this goal, you can use the following blocks for modeling the behaviour of the device:



They simulate an integration or derivative as done by an on-board computer

Discrete Worlds - Actuators

The desired control is actuated by a physical device. Therefore, the desired control will differ from the one commanded to the system.

Actuators can be seen as converters from the discrete world to the continuous world!

To model the behaviour of the physical actuator, you can use for example:



Rate Limiter

introduces a transient. Should be set in **continuous mode**!

This is just a simple way to model an actuator. There are many more blocks to use according to the actuator you have and the behavior you want to associate to your actuator!

Useful resources



Simulink Onramp: a course on Simulink by the people who invented it